Lab 2: Selecting a Software Process Model

This is a decision-making experiment - not a programming experiment.

Recommended tool: Microsoft Word
Alternative: Google Docs / LibreOffice Writer

What will you create?

- For each case: identify project characteristics.
- Compare the five process models.
- Select the most suitable model.
- Write 2-3 reasons explaining the selection.
- Finish with a short conclusion/result.

Which software/tool should students use?

Microsoft Word is enough for this experiment.

- Why Word? The output is mainly analysis, comparison tables and written justification.
- No coding is required.
- No UML tool is required in Lab 2.
- Students may use Google Docs if Word is not available.
- Save the final file as .docx or PDF according to your submission rule.

Before starting in Word - create the document

- Open Microsoft Word → Blank document.
- Type heading: SOFTWARE ENGINEERING LAB - Experiment 2.
- Type Name, Roll No., Date and Class/Section.
- Add Aim: “To compare software process models and select a suitable model for different software projects.”
- Create a section called “Process Model Comparison”.

Step 1 - Make the comparison table

Factor	Waterfall	Incremental	Prototype	Spiral	Agile
Requirements	Stable	Partly stable	Unclear	Can change	Changing
Early delivery	Low	Yes	Sample	Possible	Yes
Customer feedback	Low	Medium	High	High	Very high
Risk handling	Low	Medium	Medium	Very high	Medium

Students can reproduce this table in Word using Insert → Table.

Step 2 - Analyse Case 1

CASE 1: College Payroll System

Salary rules, allowances and deductions are already known. Major changes are not expected during development.

- Requirements
- Risk
- Frequent customer feedback
- Possible choice
- Reason

Step 3 - Analyse Case 2

CASE 2: New Food-Delivery App

The startup wants frequent releases. Users may request new features after trying the app.

- Requirements
- Early working software
- Customer feedback
- Possible choice
- Reason

Step 4 - Analyse Case 3

CASE 3: Hospital Emergency Monitoring System

Failure can have serious consequences. Technical and safety risks must be studied carefully.

- Risk
- System
- Risk analysis
- Possible choice
- Reason

Step 5 - Analyse Case 4

CASE 4: New College Portal Interface

Students and staff are not sure how the new screens should look. They want to see sample screens before finalising requirements.

- Requirements
- Possible choice
- Reason